Homework 09 - Transfer functions

Deadline: 3rd test date (21th of May)
The mandatory questions account for 100 points, while the optional questions together contribute up to an additional 20 points.

1. Explain the relationship between filters and linear time invariant systems.
2. Explain the stability criterion for digital filters in the z-plane. What is the significance of the unit circle in this context?
3. What is the relationship between the s-plane and the z-plane when analyzing digital filters? Explain the mapping from continuous to discrete systems.
4. Explain the significance of poles and zeros in the context of the z-transform. How do they influence the behavior of digital filters? How can it be used to analyze the stability and frequency response of digital filters?
5. Describe the frequency response of a digital filter. How is it typically represented, and what information does it provide?
6. How does the Bode plot for discrete-time systems differ from that for continuous-time systems?
7. For the transfer function $H(z) = \frac{1}{z-0.5}$, determine whether the system is stable and find the corresponding impulse response sequence.
8. Study the stability of the system with transfer function
$$H(z) = \frac{1}{z^2 - z + 0.5}$$
and determine its impulse response.
9. A processor has a transfer function $\frac{0.4}{z-0.5}$. Find the response to a discrete unit step (first four terms only).
10. Write a Python script to plot the pole-zero diagram for the transfer function $H(z) = \frac{z^2+0.5z+1}{z^2-1.5z+0.7}$. Analyze the stability based on the pole locations.
11. Using the `numpy.roots` function write a script to plot the pole-zero diagram for a given transfer function $H(z) = \frac{z^2+0.5z+1}{z^2-1.5z+0.7}$. Analyze the stability based on the pole locations.
12. Given the transfer function $H(z) = \frac{z+0.5}{z^2-0.7z+0.1}$, compute the first five terms of the impulse response using the inverse z-transform.
13. Define FIR and IIR filters. Explain the main differences between them in terms of complexity, efficiency, stability and phase response.
14. Given the transfer function $H(z) = \frac{z+1}{z-0.5}$, calculate the frequency response at 0 Hz, 1 Hz, and 2 Hz if the sampling frequency $f_s = 8$ Hz. Provide both gain and phase angle for each frequency.
15. For the transfer function $H(z) = \frac{1-0.8z^{-1}}{1-0.5z^{-1}}$, determine the frequency response at 0 Hz, 1 Hz, and 2 Hz if the sampling frequency $f_s = 10$ Hz. Provide both gain and phase angle for each frequency.